

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q23-2293

Weslaco 230 kV
May 02, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2293. The Interconnection Customer proposes a 1475.2 MW natural gas combined cycle facility interconnecting at the Weslaco 230 kV substation in the Illinois area. The request was received on March 08, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 414 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$49,207,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Estill Springs - Weslaco 230.0 kV	Weslco - Commonwealth Edison Des Plaines 230.0 kV Line	196.6	430	42.9
Weslaco - Commonwealth Edison Des Plaines 230.0 kV	Estill Springs - Weslaco 230.0 kV	154.9	510	38.1
Hermiston - Weslaco 230.0 kV	Weslaco - Estill Springs 230.0 kV	122.5	585	23.5
Jones Chapel - Bluebonnet 230.0 kV	Russell Fork - Robison Park 230.0 kV	110.7	1,701	24.4
Hermiston Substation - Denton West Interchange 230.0 kV line	Weslaco - Estill Springs 230.0 kV line	103.5	515	21.9
Big Oaks - Laureles Substation 100.0 kV Line	Dekalb - Jones Chapel 500.0 kV	101.9	396	5.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Weslaco - Estil Springs 230.0 kV	Rebuild line with higher-capacity conductor	\$16,713,000	79% (\$13,253,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Commonwealth Edison Des Plaines Substation - Weslaco 230.0 kV	Rebuild line with higher-capacity conductor	\$7,142,000	79% (\$5,666,000)
Hermiston - Weslaco 230.0 kV	Rebuild line with higher-capacity conductor	\$32,603,000	79% (\$25,690,000)
Jones Chapel - Bluebonnet 230.0 kV	Rebuild line with higher-capacity conductor	\$24,701,000	11% (\$2,794,000)
Hermiston Substation - Denton West Interchange 230.0 kV line	Reconductor line	\$2,134,000	79% (\$1,680,000)
Big Oaks - Laureles Substation 100.0 kV Line	Reconductor line	\$1,005,000	12% (\$124,000)
	Total allocated to project		\$49,207,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.